CHAPTER B

Fluency bank

Two questions per session, in 6 to 8 minutes including the correction. The identifiers allow revisits; they are not a list to be done in full at every lesson. The availability dates are based on the planned teaching, and should be shifted if the topic has not yet been studied.

R001 – Fractions

Calculate $\frac{2}{3} + \frac{1}{4}$. **Solution.** $\frac{11}{12}$: common denominator 12.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R002 – Order of operations

Calculate $5 - 2(3 - 7)$. **Solution.** 13: brackets, product, difference.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R003 – Powers

Simplify $\frac{10^5 \times 10^{-3}}{10^4}$. **Solution.** $10^{-2} = 0.01$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R004 – Roots

Simplify $\sqrt{50}$. **Solution.** $5\sqrt{2}$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R005 – Proportion

18 represents 30% of a total. What is the total? **Solution.** $\frac{18}{0.3} = 60$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R006 – Direct rate

A price of 80 falls by 15%. Final value? **Solution.** $80 \times 0.85 = 68$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R007 – Initial value

After a 20% increase, the value is 72. Initial value? **Solution.** $\dfrac{72}{1.2} = 60$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R008 – Calculated rate

A quantity goes from 40 to 50. Rate? **Solution.** $\dfrac{50-40}{40} = 25\%$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R009 – Successive rates

A 20% increase then a 20% decrease: overall rate? **Solution.** $1.2 \times 0.8 = 0.96$: a 4% decrease.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R010 – Reciprocal rate

Which rate compensates a 20% decrease? **Solution.** $\dfrac{1}{0.8} = 1.25$: a 25% increase.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R011 – Zero product

Solve $(2x-1)(x+3) = 0$. **Solution.** $x = \dfrac{1}{2}$ or $x = -3$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R012 – Expansion

Expand $(2x-3)^2$. **Solution.** $4x^2 - 12x + 9$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R013 – Factorisation

Factorise $3x^2 - 12x$. **Solution.** $3x(x-4)$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R014 – Sign of an affine function

Solve $-3x + 6 \geq 0$. **Solution.** $x \leq 2$: reverse the inequality when dividing by -3.
Prerequisites: Previous learning
Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R015 – Slope

Slope of the line through $(1, 4)$ and $(3, 0)$? **Solution.** -2.
Prerequisites: Previous learning
Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R016 – Line

Equation of the line of slope -2 through $(1, 4)$? **Solution.** $y = -2x + 6$.
Prerequisites: Previous learning
Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R017 – Weighted mean

The values 2 and 5 have counts 3 and 2. Mean? **Solution.** $\dfrac{6 + 10}{5} = 3.2$.
Prerequisites: Previous learning
Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R018 – Median

Median of 1, 2, 2, 5, 8, 10? **Solution.** $\dfrac{2 + 5}{2} = 3.5$ using the convention of the mean of the two central values.
Prerequisites: Previous learning
Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R019 – Quartiles

For the data set 1, 2, 3, 4, 5, 6, 7, 8, give Q_1 and Q_3 with the ceiling ranks $\left\lceil \dfrac{n}{4} \right\rceil$ and $\left\lceil \dfrac{3n}{4} \right\rceil$. **Solution.** $Q_1 = 2$, $Q_3 = 6$.
Prerequisites: Previous learning
Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R020 – Standard deviation of data

For 1 and 3 with equal weights, mean, variance and standard deviation? **Solution.** Mean 2, variance $\dfrac{[(1-2)^2 + (3-2)^2]}{2} = 1$, standard deviation 1.
Prerequisites: Previous learning
Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R021 – Units

Convert 1 min 40 s to seconds; 2400 frames over this duration give how many per second? **Solution.** 100 s; 24 frames/s.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R022 – Simple probability

With a fair die, probability of a multiple of 3? **Solution.** $\frac{2}{6} = \frac{1}{3}$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R023 – Complement

$P(A) = \frac{3}{8}$. Give $P(\overline{A})$. **Solution.** $\frac{5}{8}$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R024 – Interval

Write $x \geq -2$ and $x < 3$ as an interval. **Solution.** $[-2, 3)$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R025 – Logic

Negate “all the students passed. **Solution.** “At least one student did not pass.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R026 – Histogram

Two classes of widths 2 and 4 have the same count 20. In a histogram with areas proportional to counts, what are the relative heights? **Solution.** Heights 10 and 5: the area, not the height alone, encodes the count.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R027 – Pie chart

A sector measures 72°. Proportion? **Solution.** $\frac{72}{360} = 20\%$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R028 – Scale

An axis starts at 90. Bars stop at 100 and 110. Is the second value double the first? **Solution.** No: $\frac{110}{100} = 1.1$. The heights drawn above 90 make this comparison misleading.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R029 – Percentage of a percentage

40% of the students take A; among them, 25% take B. What proportion takes A and B? **Solution.** $0.4 \times 0.25 = 10\%$.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R030 – Proportion and count

A category represents 0.125 of 80 people. Count? **Solution.** 10 people.

Prerequisites: Previous learning

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.

R031 – Defining and representing a sequence

Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; suggest an explicit formula. **Solution.** $2, 3, 6, 11$. The formula $w_n = n^2 + 2$ works: it gives the first term and $(n+1)^2 + 2 - (n^2 + 2) = 2n + 1$.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-09; suggested revisits: 2026-09-17, 2026-10-01, 2026-11-03.

R032 – Arithmetic sequences

We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine r and u_0. **Solution.** $37 - 17 = 5r$, so $r = 4$ and then $u_0 = 17 - 3 \times 4 = 5$.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-11; suggested revisits: 2026-09-22, 2026-10-06, 2026-11-03.

R033 – Geometric sequences

Study the variation of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$. **Solution.** $u_{n+1} - u_n = -4 \times 3^n < 0$: decreasing. v_n alternates between 2 and -2: neither increasing nor decreasing.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-16; suggested revisits: 2026-09-24, 2026-10-08, 2026-11-05.

R034 – Sums, indices and proofs

Calculate $\sum_{k=2}^{5} 2^k$ without the summation symbol in the answer. **Solution.** $4+8+16+32=60$, or $\frac{4(1-2^4)}{1-2}=60$.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-18; suggested revisits: 2026-09-29, 2026-10-13, 2026-11-10.

R035 – Conjecturing a limit and programming a threshold

For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$. **Solution.** $u_3 = 6$: index 3 for ≤ 6, index 4 for < 6.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine function, Python loop.

Available from 2026-09-18; suggested revisits: 2026-09-29, 2026-10-13, 2026-11-10.

R036 – Three forms, three readings

Give three forms of $g(x) = x^2 + 2x - 3$. Read off its roots and its minimum. **Solution.** $g(x) = (x+1)^2 - 4 = (x-1)(x+3)$. Roots 1 and -3, minimum -4 at -1.

Prerequisites: Special products, zero-product property, sign of a product, square roots.

Available from 2026-09-23; suggested revisits: 2026-10-01, 2026-10-15, 2026-11-12.

R037 – Discriminant and solving

Solve $3x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$. **Solution.** The first is $3(x-1)^2 = 0$, so 1. For the second, $\Delta = -16 < 0$: no real solution.

Prerequisites: Special products, zero-product property, sign of a product, square roots.

Available from 2026-09-25; suggested revisits: 2026-10-06, 2026-11-03, 2026-11-17.

R038 – Factorising without automating the discriminant

Determine the polynomial f of degree 2 such that $f(1) = f(4) = 0$ and $f(0) = -8$. **Solution.** $f(x) = a(x-1)(x-4)$; $4a = -8$, so $f(x) = -2(x-1)(x-4)$.

Prerequisites: Special products, zero-product property, sign of a product, square roots.

Available from 2026-09-30; suggested revisits: 2026-10-08, 2026-11-03, 2026-11-19.

R039 – Sign, inequalities and optimisation

Solve $x^2 - 4x - 5 < 0$, then maximise $x(8-x)$ on $[0,3]$. **Solution.** $(x-5)(x+1) < 0$ gives $(-1,5)$. The vertex of $x(8-x)$ is at 4, outside the domain; on $[0,3]$ the function is increasing and its maximum is 15 at 3.

Prerequisites: Special products, zero-product property, sign of a product, square roots.

Available from 2026-10-02; suggested revisits: 2026-10-13, 2026-11-03, 2026-11-24.

R040 – Projection and angle

Two vectors have norms 2 and 5 and form 90°, then 180°. Calculate the products. **Solution.** 0 then -10. The cosine is 0 and -1 respectively.

Prerequisites: Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

Available from 2026-10-07; suggested revisits: 2026-10-15, 2026-11-03, 2026-11-26.

R041 – Coordinates, norm and orthogonality

For which real number t are the vectors $(t, 2)$ and $(3, -6)$ orthogonal? **Solution.** $3t - 12 = 0$, so $t = 4$.

Prerequisites: Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

Available from 2026-10-09; suggested revisits: 2026-11-03, 2026-12-01.

R042 – Vector calculus and identities

Given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$. **Solution.** The square is $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.

Prerequisites: Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

Available from 2026-10-14; suggested revisits: 2026-11-03, 2026-11-05, 2026-12-03.

R043 – Al-Kashi and choosing a method

A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine the angle A. **Solution.** $32 = 16 + 16 - 32\cos A$, so $\cos A = 0$ and $A = 90°$.

Prerequisites: Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

Available from 2026-10-16; suggested revisits: 2026-11-03, 2026-11-10, 2026-12-08.

R044 – The circle with diameter AB

For $A(0,0)$, $B(6,0)$, determine the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$. **Solution.** $MI^2 - 9 = 7$: circle with centre $(3,0)$ and radius 4.

Prerequisites: Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

Available from 2026-10-16; suggested revisits: 2026-11-03, 2026-11-10, 2026-12-08.

R045 – Rate of change and secants

Calculate the rate of $f(x) = x^2 + 3x$ between 2 and $2 + h$. **Solution.** $f(2+h) - f(2) = 7h + h^2$; rate $7 + h$ for $h \neq 0$.

Prerequisites: Images, slope of a line, algebraic manipulation, square root and absolute value.

Available from 2026-11-04; suggested revisits: 2026-11-12, 2026-11-26, 2027-01-05.

R046 – Derivative at a point

For $f(x) = 3x^2 - 2x$, calculate $f'(1)$ from the rate. **Solution.** The rate is $4 + 3h$, so the derivative at 1 is 4.

Prerequisites: Images, slope of a line, algebraic manipulation, square root and absolute value.

Available from 2026-11-06; suggested revisits: 2026-11-17, 2026-12-01, 2027-01-05.

R047 – Tangent and affine approximation

Given $f(3) = 5$ and $f'(3) = -2$. Write the tangent and approximate $f(3.04)$. **Solution.** $y = 5 - 2(x - 3) = -2x + 11$; approximation $5 - 2(0.04) = 4.92$.

Prerequisites: Images, slope of a line, algebraic manipulation, square root and absolute value.

Available from 2026-11-11; suggested revisits: 2026-11-19, 2026-12-03, 2027-01-05.

R048 – When differentiability fails

Justify that $g(x) = |x - 2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \sqrt{x}$ at 0 through its rate. **Solution.** The rate at 2 is $\dfrac{|h|}{h}$, with two opposite limiting values. For the square root at 0, the rate $\dfrac{1}{\sqrt{h}}$ for $h > 0$ exceeds every bound as h gets closer to 0: no real derivative.

Prerequisites: Images, slope of a line, algebraic manipulation, square root and absolute value.

Available from 2026-11-13; suggested revisits: 2026-11-24, 2026-12-08, 2027-01-05.

R049 – Conditioning: changing the denominator

Out of 60 objects, 20 are red; among the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$. **Solution.** $\dfrac{5}{60} = \dfrac{1}{12}$ and $\dfrac{5}{20} = \dfrac{1}{4}$.

Prerequisites: Fractions, proportions, probabilities on a finite sample space, intersection and union.

Available from 2026-11-18; suggested revisits: 2026-11-26, 2026-12-10, 2027-01-07.

R050 – Trees, partitions and total probability

$P(A) = \dfrac{1}{3}$, $P_A(B) = \dfrac{3}{4}$, $P_{\overline{A}}(B) = \dfrac{1}{4}$. Calculate $P(B)$. **Solution.** $\dfrac{1}{4} + \dfrac{1}{6} = \dfrac{5}{12}$.

Prerequisites: Fractions, proportions, probabilities on a finite sample space, intersection and union.

Available from 2026-11-20; suggested revisits: 2026-12-01, 2026-12-15, 2027-01-12.

R051 – Independence and disjointness

Given $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.18$. Are they independent? **Solution.** No: $0.4 \times 0.5 = 0.20 \neq 0.18$.

Prerequisites: Fractions, proportions, probabilities on a finite sample space, intersection and union.

Available from 2026-11-25; suggested revisits: 2026-12-03, 2026-12-17, 2027-01-19.

R052 – Successive trials and Bernoulli trials

Four independent trials, each with probability $p = \frac{1}{3}$. Calculate the probability of four failures, then of at least one success. **Solution.** $(\frac{2}{3})^4 = \frac{16}{81}$ then $1 - \frac{16}{81} = \frac{65}{81}$.

Prerequisites: Fractions, proportions, probabilities on a finite sample space, intersection and union.

Available from 2026-11-27; suggested revisits: 2026-12-08, 2027-01-05, 2027-01-19.

R053 – Simulating and discussing the model

A simulation places 2000 points at random, uniformly, in the square $[0,1] \times [0,1]$; 675 lie below the curve with equation $y = x^2$. What area does it estimate? **Solution.** $\frac{675}{2000} = 0.3375$; this is an estimate, not the exact value.

Prerequisites: Fractions, proportions, probabilities on a finite sample space, intersection and union.

Available from 2026-11-27; suggested revisits: 2026-12-08, 2027-01-05, 2027-01-19.

R054 – Derivative function and standard functions

Differentiate x^5, x^{-2} and $\sqrt{x}$, stating the domains. **Solution.** $5x^4$ on $\mathbb{R}$; $-\frac{2}{x^3}$ for $x \neq 0$; $\frac{1}{2\sqrt{x}}$ for $x > 0$.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-02; suggested revisits: 2026-12-10, 2027-01-05, 2027-01-21.

R055 – Sum, product, reciprocal and quotient

Differentiate $(x^2+1)(3x-2)$, then $\frac{1}{x^2+1}$. **Solution.** $2x(3x-2) + 3(x^2+1) = 9x^2 - 4x + 3$; $-\frac{2x}{(x^2+1)^2}$, on $\mathbb{R}$.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-04; suggested revisits: 2026-12-15, 2027-01-05, 2027-01-26.

R056 – Sign of the derivative and extrema

Study $f(x) = x^2 - 4x + 1$ on $[0,3]$ and give its extrema on this interval. **Solution.** $f' = 2x - 4$, so decreasing up to 2 then increasing. Minimum -3 at 2; values at the endpoints 1 and -2, so maximum 1 at 0.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-09; suggested revisits: 2026-12-17, 2027-01-05, 2027-01-28.

R057 – Parity, inequalities and comparison

Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$. **Solution.** First even, second odd; difference $x^2 + 1 - 2x = (x-1)^2 \geq 0$, equality at 1.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-11; suggested revisits: 2027-01-05, 2027-02-02.

R058 – Optimisation

A rectangle of perimeter 12 m has width x. Determine the maximum area by differentiation. **Solution.** $A(x) = x(6-x)$ on $[0,6]$; $A\prime = 6-2x$. Maximum 9 m^2 at 3.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-11; suggested revisits: 2027-01-05, 2027-02-02.

R059 – Normal vector and Cartesian equation

Equation of the line through $(2,-1)$ with normal $(3,4)$. **Solution.** $3(x-2)+4(y+1)=0$, that is $3x+4y-2=0$.

Prerequisites: Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.

Available from 2027-01-06; suggested revisits: 2027-01-19, 2027-01-28, 2027-03-02.

R060 – Orthogonal projection and distance

Project $P(4,1)$ onto $d : x = 2$. **Solution.** $H(2,1)$ and distance 2; the normal is horizontal.

Prerequisites: Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.

Available from 2027-01-08; suggested revisits: 2027-01-19, 2027-02-02, 2027-03-02.

R061 – Equation of a circle

Recognise $x^2+y^2+2x-8y+17=0$. **Solution.** $(x+1)^2+(y-4)^2=0$: the single point $(-1,4)$, not a circle of strictly positive radius.

Prerequisites: Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.

Available from 2027-01-13; suggested revisits: 2027-01-21, 2027-02-04, 2027-03-04.

R062 – Solving a configuration

Intersect $(x-1)^2+(y+2)^2=9$ with $y=1$. **Solution.** $(x-1)^2=0$, so the only point is $(1,1)$; the line is tangent to the circle.

Prerequisites: Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.

Available from 2027-01-15; suggested revisits: 2027-01-26, 2027-02-09, 2027-03-09.

R063 – Variable, event and distribution

Two fair coins are tossed. X counts the heads. Write its distribution and $P(X \leq 1)$. **Solution.** Values 0, 1, 2; probabilities $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$. $P(X \leq 1) = \frac{3}{4}$.

Prerequisites: Probability, weighted mean, squares, trees and lists.

Available from 2027-01-20; suggested revisits: 2027-01-28, 2027-02-11, 2027-03-11.

R064 – Expectation and linearity

A gross gain is 10 with probability $\frac{1}{5}$ and 0 otherwise. What stake makes the game fair? **Solution.** $E(G) = \frac{10}{5} = 2$. For a stake m, $E(G - m) = 2 - m$; fair stake 2.

Prerequisites: Probability, weighted mean, squares, trees and lists.

Available from 2027-01-22; suggested revisits: 2027-02-02, 2027-03-02, 2027-03-16.

R065 – Variance, standard deviation and König–Huygens

For X with values $-1, 1$ equally likely, calculate the expectation, variance and standard deviation. **Solution.** $E(X) = 0$, $E(X^2) = 1$, variance 1 and standard deviation 1.

Prerequisites: Probability, weighted mean, squares, trees and lists.

Available from 2027-01-27; suggested revisits: 2027-02-04, 2027-03-02, 2027-03-18.

R066 – Transforming and comparing gains

$E(X) = 3$, $V(X) = 4$. Give the characteristics of $Y = -2X + 5$. **Solution.** $E(Y) = -1$, $V(Y) = 16$, $\sigma(Y) = 4$.

Prerequisites: Probability, weighted mean, squares, trees and lists.

Available from 2027-01-29; suggested revisits: 2027-02-09, 2027-03-02, 2027-03-23.

R067 – Samples and observed mean

X takes the values 0 and 2 with equal probability. Calculate the observed mean of the sample 2, 0, 2, 2, 0, then its gap from $E(X)$. **Solution.** $m = \frac{6}{5} = 1.2$ and $E(X) = 1$: the gap is 0.2 for this sample.

Prerequisites: Probability, weighted mean, squares, trees and lists.

Available from 2027-01-29; suggested revisits: 2027-02-09, 2027-03-02, 2027-03-23.

R068 – Defining a new function

What is the equation of the tangent to e^x at 1? **Solution.** $y = e + e(x - 1) = ex$.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-03; suggested revisits: 2027-02-11, 2027-03-02, 2027-03-25.

R069 – Algebraic properties and comparisons

Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$. **Solution.** e^{2x+1}; $x + 2 > 2x$, so $x < 2$.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-05; suggested revisits: 2027-03-02, 2027-03-30.

R070 – Differentiating and studying

Differentiate $g(t) = 3e^{-2t}$ and study its direction of variation. **Solution.** $g'(t) = -6e^{-2t} < 0$ on $\mathbb{R}$: strictly decreasing.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-10; suggested revisits: 2027-03-02, 2027-03-04, 2027-04-06.

R071 – From discrete to continuous

For $Q(t) = 5e^{0.3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$. **Solution.** $u_n = 5(e^{0.3})^n$; the ratio over two units is $e^{0.6}$.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-12; suggested revisits: 2027-03-02, 2027-03-09, 2027-04-06.

R072 – Building numerically and proving an inequality

Show that $e^{2x} \geq 1 + 2x$ for every real number x. **Solution.** Apply the inequality proved with the argument $2x$; equality only at x=0.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-12; suggested revisits: 2027-03-02, 2027-03-09, 2027-04-06.

R073 – Measuring an angle in radians

Convert 45°, 120°; give the arc for radius 2 m and angle $\dfrac{3\pi}{4}$. **Solution.** $\dfrac{\pi}{4}$, $\dfrac{2\pi}{3}$; length $\dfrac{3\pi}{2}$ m.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-03; suggested revisits: 2027-03-11, 2027-03-25, 2027-04-27.

R074 – Wrapping and the image of a real number

Place the images of $\dfrac{7\pi}{2}$ and $-\dfrac{5\pi}{3}$ by reducing them to within one turn. **Solution.** $\dfrac{7\pi}{2}$ has the image of $\dfrac{3\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-05; suggested revisits: 2027-03-16, 2027-03-30, 2027-04-27.

R075 – Sine, cosine and standard values

If $\cos x = \dfrac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$. **Solution.** $\sin^2 x = \dfrac{16}{25}$; the sine is negative, so $-\dfrac{4}{5}$.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-10; suggested revisits: 2027-03-18, 2027-04-06, 2027-04-29.

R076 – Related angles and symmetries

Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{7\pi}{4}$. **Solution.** $(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$ and $(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-12; suggested revisits: 2027-03-23, 2027-04-06, 2027-05-04.

R077 – Approximating pi and putting it to use

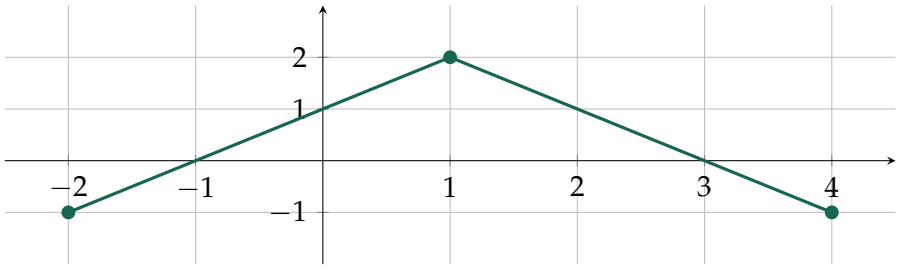
For n=6 and c=1, calculate the new chord after doubling. **Solution.** $h = \frac{\sqrt{3}}{2}$; new chord $\frac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-12; suggested revisits: 2027-03-23, 2027-04-06, 2027-05-04.

R078 – Reading a graph

On the figure R078, read $f(1)$, solve $f(x) = 0$ and $f(x) > 0$.



Solution. $f(1) = 2$; zeros -1 and 3; positive sign on $(-1, 3)$.

Prerequisites: Plotting points

Available from 2026-09-03; suggested revisits: 2026-09-10, 2026-09-24, 2026-11-03.